COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	8/22/26	**DATE SUBMITTED:**	8/24/26
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ACTIVITY NO. 4.1 Running Elevated Ad hoc Commands

1. PROCEDURE OUTPUT

Task 1: Run elevated ad hoc commands

1.

What is the result of the command? Is it successful?

- Unsuccessful

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible -i inventory all -m apt
 -a update_cache=true
node1 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "msg": "Failed to lock apt for exclusive operation: Failed to lock directory
 /var/lib/apt/lists/: E:Could not open lock file /var/lib/apt/lists/lock - open
(13: Permission denied)"
}
node2 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "msg": "Failed to lock apt for exclusive operation: Failed to lock directory
 /var/lib/apt/lists/: E:Could not open lock file /var/lib/apt/lists/lock - open
(13: Permission denied)"
}
```

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible -i inventory all -m apt
 -a update_cache=true --become --ask-become-pass
BECOME password:
node1 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_update_time": 1787576069,
    "cache_updated": true,
    "changed": true
}
node2 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_update_time": 1787576069,
    "cache_updated": true,
    "changed": true
}
```

2.

a.

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible -i inventory all -a "wh
ich vim"
node1 | CHANGED | rc=0 >>
/usr/bin/vim
node2 | CHANGED | rc=0 >>
/usr/bin/vim
```

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible -i inventory all -a "ap
t search vim-nox"
node1 | CHANGED | rc=0 >>
Sorting...
Full Text Search...
vim-nox/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.19 amd64 [installed
]
  Vi IMproved - enhanced vi editor - with scripting languages support

vim-tiny/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.19 amd64 [installe
d,automatic]
  Vi IMproved - enhanced vi editor - compact version
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.
node2 | CHANGED | rc=0 >>
Sorting...
Full Text Search...
vim-nox/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.19 amd64 [installed
]
  Vi IMproved - enhanced vi editor - with scripting languages support

vim-tiny/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.19 amd64 [installe
d,automatic]
  Vi IMproved - enhanced vi editor - compact version
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.
```

b.

```
villareal@villareal-ansible1:/var/log$ sudo cat /var/log/apt/history.log
[sudo] password for villareal:

Start-Date: 2026-02-10  00:20:26
Commandline: apt-get -y --fix-policy install
Install: libgpg-error-l10n:amd64 (1.47-3build2, automatic), e2fsprogs-l10n:amd64
 (1.47.0-2.4~exp1ubuntu4, automatic), libgpm2:amd64 (1.20.7-11, automatic), psmi
sc:amd64 (23.7-1build1, automatic), uuid-runtime:amd64 (2.39.3-9ubuntu6, automat
ic), bash-completion:amd64 (1:2.11-8, automatic), bsdextrautils:amd64 (2.39.3-9u
buntu6, automatic)
End-Date: 2026-02-10  00:20:28

Start-Date: 2026-02-10  00:21:30
Commandline: apt-get --yes -oDebug::pkgDepCache::AutoInstall=yes --force-yes upg
rade
Upgrade: libgpg-error-l10n:amd64 (1.47-3build2, 1.47-3build2.1), dpkg:amd64 (1.2
2.6ubuntu6, 1.22.6ubuntu6.5), libsmartcols1:amd64 (2.39.3-9ubuntu6, 2.39.3-9ubun
tu6.4), libpam-runtime:amd64 (1.5.3-5ubuntu5, 1.5.3-5ubuntu5.5), udev:amd64 (255
.4-1ubuntu8, 255.4-1ubuntu8.12), coreutils:amd64 (9.4-3ubuntu6, 9.4-3ubuntu6.1),
 krb5-locales:amd64 (1.20.1-6ubuntu2, 1.20.1-6ubuntu2.6), python3.12:amd64 (3.12
```

- This contains the history of apt commands and all its output

3.

a. It's a successful command and outputs that snapd has been installed to the remote servers

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible -i inventory all -m apt
 -a name=snapd --become --ask-become-pass
BECOME password:
node1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_update_time": 1787576069,
    "cache_updated": false,
    "changed": false
}
node2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_update_time": 1787576069,
    "cache_updated": false,
    "changed": false
}
```

b. Update

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible -i inventory all -m apt
 -a "name=snapd state=latest" --become --ask-become-pass
BECOME password:
node1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_update_time": 1787576069,
    "cache_updated": false,
    "changed": false
}
node2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_update_time": 1787576069,
    "cache_updated": false,
    "changed": false
}
```

Task 2: Writing our First Playbook

1.
```
  GNU nano 7.2                                  install_apache.yml
---
- hosts: all
  become: yes
  tasks:

  - name: install apache2 package
    apt:
      name: apache2
```

2.
```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible-playbook -i inventory i
nstall_apache.yml --ask-become-pass
BECOME password:

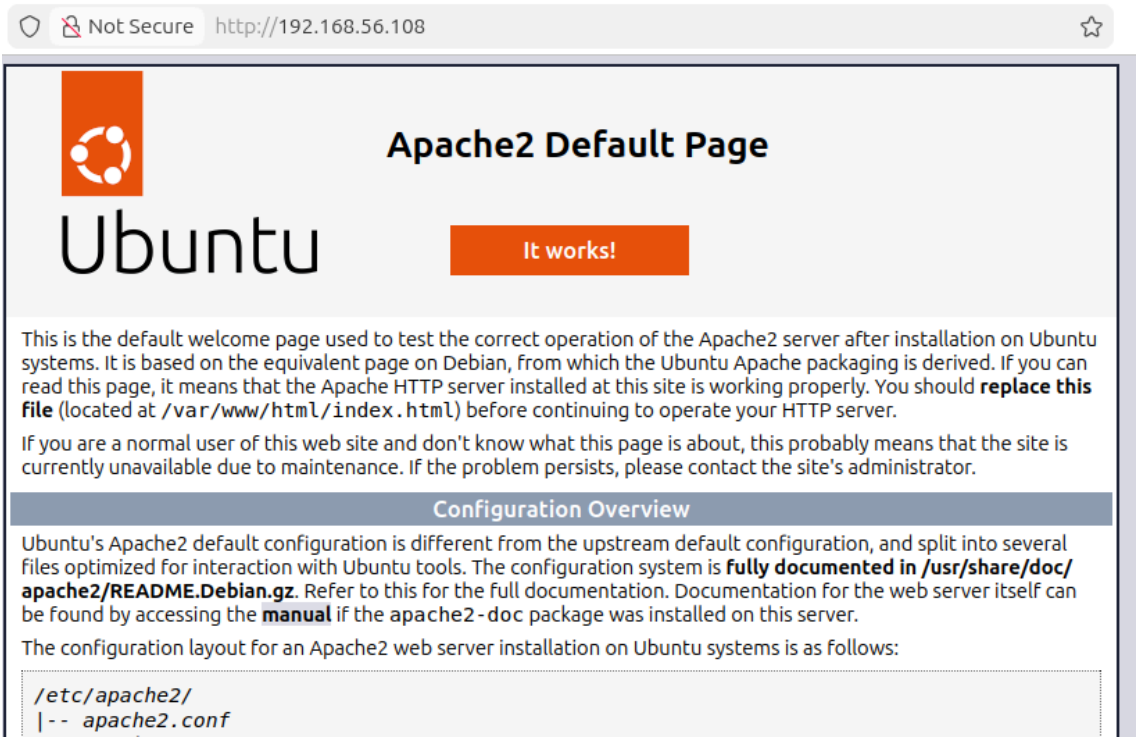
PLAY [all] *********************************************************************

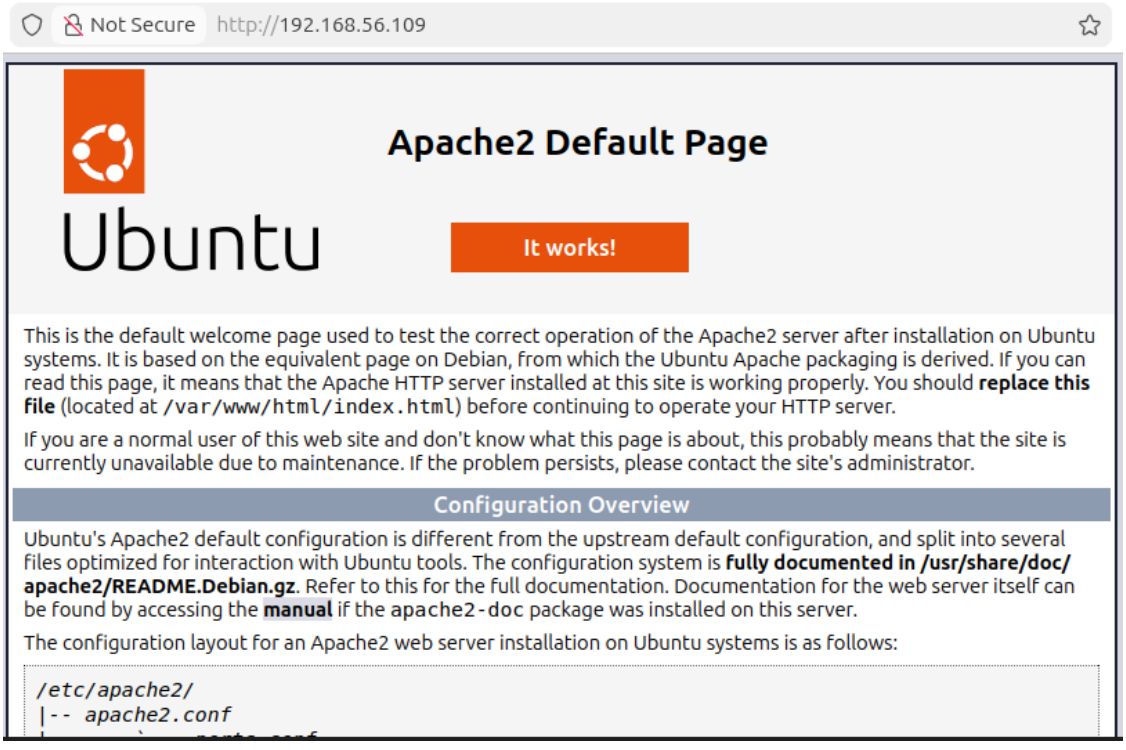
TASK [Gathering Facts] *********************************************************
ok: [node1]
ok: [node2]

TASK [install apache2 package] *************************************************
changed: [node1]
changed: [node2]

PLAY RECAP *********************************************************************
node1                      : ok=2    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
node2                      : ok=2    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

3. Proof





4. What is the output? It failed since the new changed name is unrecognizable.

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible-playbook -i inventory i
nstall_apache.yml --ask-become-pass
BECOME password:

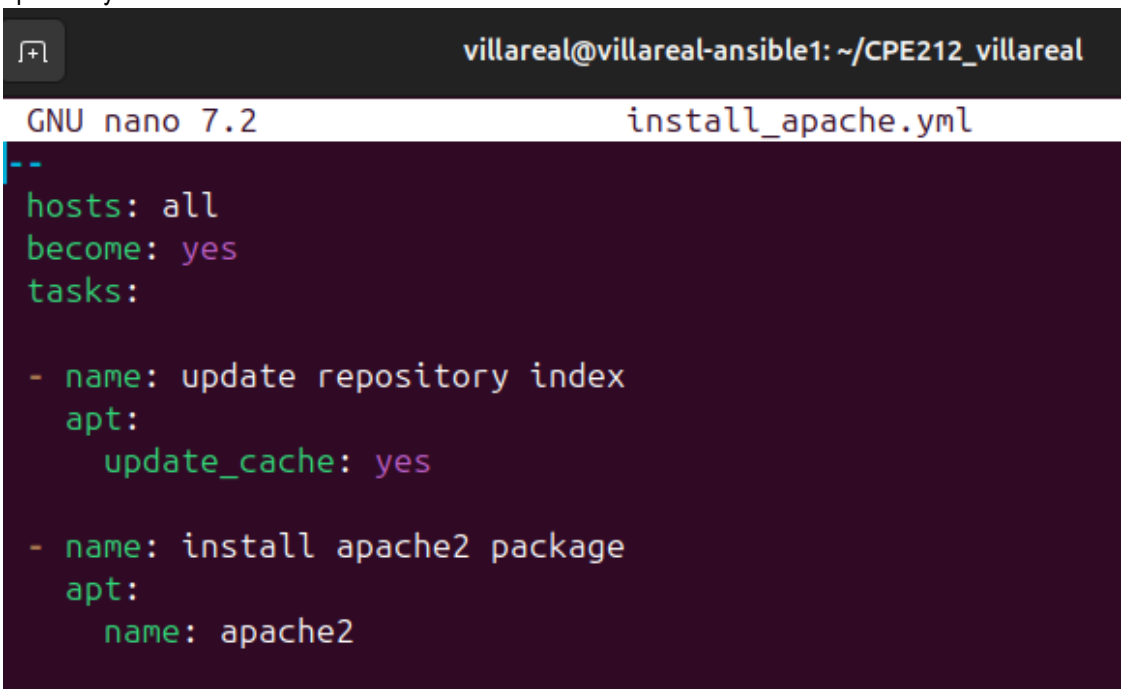
PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [node2]
ok: [node1]

TASK [install apache2 package] *************************************************
fatal: [node1]: FAILED! => {"changed": false, "msg": "No package matching 'asdas
' is available"}
fatal: [node2]: FAILED! => {"changed": false, "msg": "No package matching 'asdas
' is available"}

PLAY RECAP *********************************************************************
node1                      : ok=1    changed=0    unreachable=0    failed=1    s
kipped=0    rescued=0    ignored=0
node2                      : ok=1    changed=0    unreachable=0    failed=1    s
kipped=0    rescued=0    ignored=0
```

5. Updated .yml file



6. Did the new command change anything on the remote servers? Yes, and it outputs that the cache has been updated and the command is run.

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible-playbook -i inventory i
nstall_apache.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [node1]
ok: [node2]

TASK [update repository index] *************************************************
changed: [node1]
changed: [node2]

TASK [install apache2 package] *************************************************
ok: [node1]
ok: [node2]

PLAY RECAP *********************************************************************
node1                      : ok=3    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
node2                      : ok=3    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

7.

```
villareal@villareal-ansible1: ~/CPE212_villareal
  GNU nano 7.2                        install_apache.yml
---
- hosts: all
  become: yes
  tasks:

  - name: update repository index
    apt:
      update_cache: yes

  - name: install apache2 package
    apt:
      name: apache2

  - name: add PHP support for apache
    apt:
      name: libapache2-mod-php
```

8. Did the new command change anything on the remote servers? Yes, now there are new lines because of the changes, this would then display these outputs to the remote servers.

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible-playbook -i inventory i
nstall_apache.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [node1]
ok: [node2]

TASK [update repository index] *************************************************
changed: [node1]
changed: [node2]

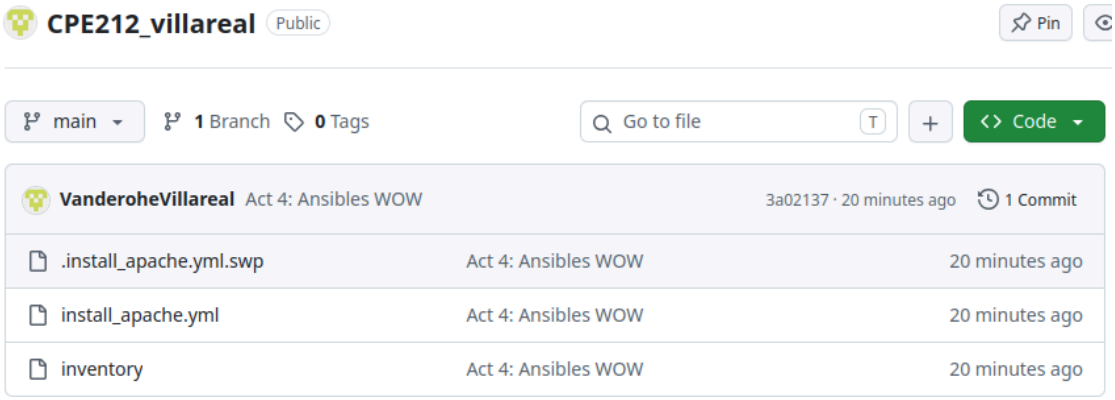
TASK [install apache2 package] *************************************************
ok: [node1]
ok: [node2]

TASK [add PHP support for apache] **********************************************
changed: [node1]
changed: [node2]

PLAY RECAP *********************************************************************
node1                      : ok=4    changed=2    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
node2                      : ok=4    changed=2    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

9. GitHub

```
villareal@villareal-ansible1:~/CPE212_villareal$ ssh -T git@github.com
The authenticity of host 'github.com (20.205.243.166)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi VanderoheVillareal! You've successfully authenticated, but GitHub does not pr
ovide shell access.
```



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What is the importance of using a playbook?
   - A playbook is important because it allows us to automate tasks on multiple remote servers. Instead of entering commands one by one, we can put the tasks in a YAML file and let Ansible execute them automatically. This makes server management faster, easier, and more organized.
2. Summarize what we have done on this activity.
   - In this activity, we used Ansible to manage two remote Ubuntu servers. We first used elevated ad hoc commands to update the package cache and install packages such as vim-nox and snapd. We also checked the installation and reviewed the APT logs. After that, we created an Ansible playbook to automatically install Apache and added additional tasks to the playbook. Finally, we synchronized our work with GitHub.

Conclusions/Learnings:

Through this activity, I learned how Ansible can make managing remote servers easier and more efficient. I learned how to use privilege escalation with –become, install packages using ad hoc commands, and create a YAML playbook for automation. I also learned how important the inventory and SSH connection are for Ansible to communicate with remote machines. Overall, the activity helped me understand how automation can reduce the need to perform repetitive server management tasks manually.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Error assistance**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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Extent of Use: **AI is used for assisting in solving the errors encountered during the activity**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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